

OptiLine

Raceline Optimisation Toolkit

Package Manual — Version 0.1.5

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Abstract

OptiLine is a Python package for computing optimal racing lines on closed tracks. It provides routines for minimum-curvature and minimum lap-time trajectory optimisation, shortest-path and optimal-shortest-path planning, cubic and clothoid spline geometry, and full kinematic profile generation (velocity, acceleration, and lap time). The package combines quadratic programming, evolutionary strategies (CMA-ES), zero-order Gaussian random methods (ZORM), direct collocation via CasADi/IPOPT, and a blackbox direct-alpha optimiser into a unified, extensible interface aimed at autonomous racing research.

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1 Introduction

Racing-line optimisation is the problem of choosing, for every point around a closed circuit, a lateral position within the drivable corridor such that a chosen cost (lap time, path curvature, path length, ...) is minimised subject to track-boundary and vehicle-dynamics constraints.

OptiLine addresses this problem at multiple levels of fidelity:

1. **Geometric methods** – minimum-curvature (QP) and optimal-shortest-path (QP) formulations that operate purely on track geometry, producing a smooth spline raceline without requiring a vehicle model.
2. **Lap-time proxy methods** – the combined curvature-and-time optimiser **Opt_min_CurvTime** uses the kinematic velocity profile as a fast surrogate for lap time and drives the search with CMA-ES or ZORM over the spline segment-length vector.
3. **Blackbox direct-alpha optimisation** – the **Blackbox_raceline** class optimises the lateral shift vector α directly, treating the kinematic lap-time oracle (or any user-supplied callable) as a blackbox cost function and using ZORM or CMA-ES for the outer search.
4. **Optimal-control methods** – a full single-track minimum-time optimal-control problem solved by direct Gauss–Legendre collocation via CasADi and IPOPT (**opt_mintime**).

The package is structured as follows:

Module	Role
solvers.py	Core optimisation solvers and raceline classes
utils.py	Spline utilities, track preprocessing, QP matrix builders
opt_mintime.py	Minimum lap-time optimal control (CasADi/IPOPT)
KinematicProfs.py	Velocity, acceleration, and lap-time profile computation

Parts of this codebase are adopted from https://github.com/TUMFTM/global_racetrajectory_optimization.

2 Installation

2.1 Requirements

OptiLine requires Python 3.10–3.12 and the following libraries:

Library	Version	Purpose
<i>numpy</i>	$\geq 1.24, < 3$	Array operations
<i>scipy</i>	$\geq 1.10, < 2$	Interpolation, linear algebra
<i>matplotlib</i>	$\geq 3.7, < 4$	Plotting
<i>quadprog</i>	$\geq 0.1.13, < 0.2$	Quadratic programming
<i>casadi</i>	$\geq 3.7, < 4$	Automatic differentiation / IPOPT

2.2 Installing with pip

```
pip install optiline
```

2.3 Installing from source (Poetry)

```
git clone https://github.com/amirali78frz/OptiLine.git
cd OptiLine
poetry install
```

2.4 Installing using a wheel file

```
pip install optiline-0.2.0-py3-none-any.whl
```

3 Package Structure

```
OptiLine/
|-- src/
|   |-- OptiLine/
|       |-- __init__.py
|       |-- solvers.py           # Optimisation algorithms
|       |-- utils.py             # Helper files
|       |-- opt_mintime.py       # Min-time optimal control
|       '-- kinematicProfs.py    # Velocity/acceleration/time profiles
|-- tests/
|   |-- General_test.py          # End-to-end geometric + proxy
|       optimisation
|   '-- General_test_optmintime.py # End-to-end min-time optimal
|       control
|-- params/
|   '-- racecar.ini              # Vehicle dynamics parameters
|-- maps/                        # Track CSV files
|-- poetry.lock
'-- pyproject.toml
```

4 Module: `utils.py`

`utils.py` provides spline mathematics, track-preprocessing helpers, and QP matrix builders used by the solvers.

4.1 `calc_splines`

```
1 coeffs_x, coeffs_y, M, normvec_normalized = calc_splines(
2     path,
3     el_lengths=None,
4     psi_s=None, psi_e=None,
5     use_dist_scaling=True,
6 )
```

Solves for curvature-continuous (C2) cubic spline coefficients through a sequence of track points. The spline passes through every point and matches heading and curvature at each knot. For a closed track, periodic boundary conditions are enforced automatically.

Parameters

Parameter	Description
path	(n+1, 2) array of (x, y) points; last row repeats first for a closed track
el_lengths	(n,) segment lengths [m]; computed from Euclidean distance if None
psi_s, psi_e	Boundary headings (rad) for the unclosed case
use_dist_scaling	Scale the spline system by segment lengths for better conditioning

Returns

Return	Description
coeffs_x	(n,4) cubic coefficients for the x -coordinate
coeffs_y	(n,4) cubic coefficients for the y -coordinate
M	(n,n) spline system matrix (used by QP solvers)
normvec_normalized	(n,2) outward unit normal vectors at each control point

4.2 interp_splines

```

1 path_interp, inds_interp, t_interp, dists_interp = interp_splines(
2     coeffs_x, coeffs_y, spline_lengths,
3     incl_last_point=False,
4     stepsize_approx=None,
5     stepsize_fixed=None,
6 )

```

Interpolates a 2-D cubic spline at a uniform arc-length step size and returns sample points, spline-segment indices, local spline parameters ($t \in [0, 1]$), and cumulative distances. Exactly one of `stepsize_approx` or `stepsize_fixed` must be supplied.

4.3 calc_spline_lengths

```

1 spline_lengths = calc_spline_lengths(
2     coeffs_x, coeffs_y,
3     quickndirty=False,
4     no_interp_points=15,
5 )

```

Numerically integrates the arc length of each cubic spline segment using Gaussian quadrature. Returns a (n,) array of segment lengths in metres.

4.4 create_raceline

```

1 (raceline, a_opt,
2  coeffs_x_opt, coeffs_y_opt,
3  spline_inds_opt, t_vals_opt,
4  s_points_opt, spline_lengths_opt,
5  el_lengths_opt) = create_raceline(
6      refline, normvec_normalized, alpha, stepsize_interp)

```

Applies lateral offsets α_i along the unit normals to the reference line, fits new cubic splines through the resulting control points, and interpolates the raceline at the requested arc-length step size.

Parameters

Parameter	Description
refline	(n,2) reference line (x, y)
normvec_normalized	(n,2) unit normals (from <code>calc_splines</code>)
alpha	(n,) lateral shift vector [m]; positive shifts toward the right boundary
stepsize_interp	Target arc-length spacing of the output raceline [m]

4.5 calc_head_curv_an

```
1 psi, kappa = calc_head_curv_an(
2     coeffs_x, coeffs_y,
3     ind_spls, t_spls,
4     calc_curv=True,
5 )
```

Analytically evaluates heading ψ and curvature κ from cubic spline coefficients at the given (segment index, local parameter) pairs.

4.6 H_f – Minimum-Curvature QP Matrices

```
1 H, f, G, h = H_f(
2     reftrack,          # (n,4): x, y, w_right, w_left
3     normvectors,      # (n,2): unit normals
4     A,                 # (n,n): spline system matrix from
      calc_splines
5     kappa_bound,      # max curvature [1/m]
6     w_veh,            # vehicle half-width [m] (safety clearance)
7     closed=True,
8 )
```

Assembles the QP matrices $(H, \mathbf{f}, G, \mathbf{h})$ for the minimum-curvature problem. H encodes the spline-curvature Hessian; G and $\mathbf{h}$ encode the track-boundary box constraints on $\boldsymbol{\alpha}$, with per-side clearance $\mathbf{w_veh}$:

$$-(w_{\text{left},i} - \delta) \leq \alpha_i \leq (w_{\text{right},i} - \delta), \quad \delta = \mathbf{w_veh}.$$

The matrices are ready for *quadprog*: `solve_qp(H, -f, -G.T, -h, 0)`.

4.7 import_veh_dyn_info

```
1 ggv, ax_max_machines = import_veh_dyn_info(
2     ggv_import_path, ax_max_machines_import_path)
```

Loads the GGV (velocity-grip-grip) diagram and the maximum machine traction table from CSV files. The GGV is an $(n_v \times 3)$ array with columns $[v, a_{x,\text{max}}, a_{y,\text{max}}]$. The traction table is an $(m \times 2)$ array with columns $[v, a_{x,\text{drive}}]$.

4.8 conv_filt

```
1 signal_out = conv_filt(signal, filt_window, closed)
```

Applies a symmetric moving-average (box-kernel) filter to a 1-D signal. `filt_window` is the number of taps (must be odd). `closed` controls whether wrap-around padding is used.

5 Module: solvers.py

solvers.py contains the high-level optimisation classes and lower-level QP-based geometric solvers.

5.1 opt_min_curv – Minimum-Curvature QP

```
1 alpha_mincurv, curv_error_max = opt_min_curv(
2     reftrack, normvectors, A,
3     kappa_bound, w_veh,
4     print_debug=False, plot_debug=False,
5     closed=True,
6 )
```

Solves a QP that minimises the integral of squared curvature of the resulting spline raceline. The optimisation variable is $\alpha \in \mathbb{R}^n$, the vector of lateral shifts applied to each reference-line point along its outward normal:

$$\mathbf{p}_i = \mathbf{r}_i + \alpha_i \hat{\mathbf{n}}_i.$$

Mathematical formulation

$$\min_{\alpha} \frac{1}{2} \alpha^\top H \alpha + \mathbf{f}^\top \alpha \quad \text{subject to} \quad G \alpha \leq \mathbf{h},$$

where H and $\mathbf{f}$ are assembled by **H_f** and the inequalities enforce track-boundary limits. Solved with *quadprog*. See [5] for a detailed derivation.

Parameters

Parameter	Description
<code>reftrack</code>	(n,4): $x, y, w_{\text{right}}, w_{\text{left}}$
<code>normvectors</code>	(n,2) unit normals from calc_splines
<code>A</code>	Spline system matrix M from calc_splines
<code>kappa_bound</code>	Maximum curvature [m^{-1}]
<code>w_veh</code>	Safety clearance per side [m]
<code>closed</code>	True for a closed circuit

Returns

Return	Description
<code>alpha_mincurv</code>	(n,) optimal lateral-shift vector [m]
<code>curv_error_max</code>	Maximum curvature residual relative to the geometric bound

5.2 OSP, ShortestPath – Optimal Shortest Path QP Matrices and Feasible Path

```

1 H, f, G, h = OSP(
2     reftrack,          # (n,4): x, y, w_right, w_left
3     normvectors,       # (n,2): unit normals
4     w_veh,             # vehicle width [m] (clearance = w_veh/2 per
                        side)
5     print_debug=False,
6 )

```

Builds the QP matrices for the *Optimal Shortest Path* problem, which minimises the sum of squared lateral displacements between consecutive raceline points:

$$\min_{\alpha} \sum_{i=0}^{n-1} \|\mathbf{p}_{i+1} - \mathbf{p}_i\|^2, \quad \mathbf{p}_i = \mathbf{r}_i + \alpha_i \hat{\mathbf{n}}_i,$$

subject to the box constraint $\alpha_i \in [-(w_{\text{left},i} - w_{\text{veh}}/2), w_{\text{right},i} - w_{\text{veh}}/2]$. Note that **w_veh** is the *full* vehicle width; the per-side clearance is **w_veh**/2. More details can be found in [1].

The matrices (**H**,**f**,**G**,**h**) are ready for *quadprog*. This function is also used internally by **Blackbox_raceline** for the 'shortest' initialisation mode (passing **w_veh** = 2*sfty so that the per-side clearance matches sfty).

```

1 raceline, alpha_sp, s_splines, vx_profile, ax_profile, kappa_opt,
   t_profile \
2     = ShortestPath(
3         reftrack, w_veh, stepsize, plot,
4         ggv_import_path, ax_max_machines_import_path)

```

Convenience wrapper: calls **OSP** to solve the shortest-path QP, then runs **create_raceline** and the full kinematic pipeline, and optionally plots the result. The function returns seven values.

Parameters

Parameter	Description
reftrack	(n,4): x, y, w _{right} , w _{left}
w_veh	Full vehicle width [m]
stepsize	Raceline interpolation step size [m]
plot	Display plots if True
ggv_import_path	Path to GGV CSV
ax_max_machines_import_path	Path to machine-traction CSV

Returns

Return	Description
raceline	(M,2) interpolated raceline [x, y]
alpha_sp	(n,) lateral-shift vector [m]
s_splines	(M,) cumulative arc-length [m]
vx_profile	(M,) velocity profile [m/s]
ax_profile	(M,) longitudinal acceleration [m/s ²]
kappa_opt	(M,) curvature profile [m ⁻¹]
t_profile	(M+1,) cumulative lap time [s]

5.3 Class Opt_min_CurvTime

Opt_min_CurvTime combines spline geometry with a kinematic velocity profile to create a fast surrogate for lap time. A meta-optimiser (CMA-ES or ZORM) searches over the spline *segment-length* vector $\mathbf{s} \in \mathbb{R}_{>0}^n$ to minimise this surrogate. For each candidate $\mathbf{s}$, the objective function **f_t** fits splines with those lengths, solves the minimum-curvature QP for α , builds the raceline, and returns the predicted lap time.

Constructor

```

1  obj = Opt_min_CurvTime(
2      reftrack,                # (n,4): x, y, w_right, w_left
3      center,                  # (n,4): centre-line (same as reftrack
   for 1-pass runs)
4      mu      = 0.05,          # ZO smoothing parameter
5      h       = 0.001,         # ZO gradient step size
6      kapb    = 0.7,           # max curvature bound [1/m]
7      sfty    = 1.0,           # safety clearance per side [m] (=
   vehicle half-width)
8      t       = 1,             # ZO directions averaged per gradient
   estimate
9      si      = 0.8,           # raceline interpolation step size [m]
10     vm      = 22.88,          # maximum velocity [m/s]
11     m_veh    = 1000.0,         # vehicle mass [kg]
12     drag_coeff = 0.0,          # drag coefficient [kg/m]
13     MC       = 1,             # Monte Carlo repetitions for ZO
14     min_s    = None,          # lower bound on segment lengths (auto if
   None)
15     max_s    = None,          # upper bound on segment lengths (auto if
   None)
16     sigma    = 0.001,         # initial CMA-ES step size
17     iterations_ZO = 300,      # ZO iteration budget
18     iterations_CMA = 30,      # CMA-ES generation budget
19     popsize   = 16,           # CMA-ES population size
20     ggv_import_path = "maps/ggv.csv",
21     ax_max_machines_import_path = "maps/ax_max_machines.csv",
22     fw        = 3,            # velocity-profile filter window length
23     refine_every = None,      # path-refinement cadence (see below)
24     refine_subsample = 1,     # sub-sampling stride after refinement
25 )

```

When `min_s` and `max_s` are both `None` (default) the bounds are set automatically from the Euclidean segment lengths of `reftrack`: $s_{\min} = \min_i \|\mathbf{r}_{i+1} - \mathbf{r}_i\|$ and $s_{\max} = 1.7 \times \max_i \|\mathbf{r}_{i+1} - \mathbf{r}_i\|$.

Path refinement (`refine_every`). When set to a positive integer q , **CurveLenOpt** splits the total iteration budget into blocks of q iterations. After each block (except the last) the current best raceline is extracted, the control points are shifted by the QP-optimal α , and the result becomes the new reference track. Track half-widths are updated so the physical boundaries stay in place:

$$w_{\text{right},i}^{\text{new}} = w_{\text{right},i} - \alpha_i, \quad w_{\text{left},i}^{\text{new}} = w_{\text{left},i} + \alpha_i.$$

Call **reset()** to restore the original track between runs.

Key methods

f_t(s) Objective function. Given segment lengths $\mathbf{s}$, fits splines, solves the curvature QP, builds the kinematic profile, and returns the predicted lap time.

CurveLenOpt(solver='Z0', refine_every=None) Runs the meta-optimisation. `solver` is 'Z0' (ZORM) or 'CMA' (CMA-ES). Returns the optimised segment-length vector `ds_opt`. When `refine_every` is set it overrides the instance default for this call only.

generate_raceline(ds=None, solver='Z0') Given an optimised segment-length vector `ds` (or runs **CurveLenOpt** if `None`), returns the interpolated raceline coordinates.

generate_kinProfs(ds=None, solver='Z0', refine_every=None) Runs the full pipeline and returns: `(s, vx, ax, kappa, t_profile, raceline)` – cumulative distance, velocity, acceleration, curvature, lap-time, and raceline coordinates.

Comparison(plot='N', output='Z0') Runs ZO, CMA-ES, minimum-curvature, and centre-line solutions, prints a lap-time comparison table, and optionally produces comparison plots. Returns the profiles of the requested `output` case.

reset() Restores `self.reftrack` to the original track supplied at construction time, discarding any in-place path refinements.

5.4 Class ConstrainedCMAES_t

A self-contained implementation of CMA-ES (Covariance Matrix Adaptation Evolution Strategy) with box-constraint handling via death penalty.

Constructor

```

1 cma = ConstrainedCMAES_t(
2     f,                # callable: objective f(x) -> float
3     mean,             # initial mean vector (n,)
4     sigma,            # initial step size (scalar)
5     popsize,          # population size lambda
6     bounds1,          # upper bounds (n,)
7     bounds2,          # lower bounds (n,)
8 )
9 s_opt = cma.optimize(iterations)
```

Algorithm outline

Each generation the algorithm:

1. Samples λ candidate solutions from $\mathcal{N}(\mathbf{m}, \sigma^2 C)$, clipping to `[bounds2, bounds1]`.
2. Evaluates f at each candidate.
3. Updates the mean $\mathbf{m}$, step-size σ , and covariance C via cumulative step-size adaptation (CSA) and rank- μ update.

See [4] for the full algorithm.

5.5 Class ZORM

Zero-Order Random Method (ZORM) is a gradient-free optimiser that estimates the gradient of a μ -smoothed surrogate using random perturbations.

Constructor

```

1  opt = ZORM(
2      func,          # callable: objective f(x) -> float
3      x0,            # initial point (n,)
4      num_iterations, # total gradient steps
5      mu = 0.05,      # smoothing / perturbation radius
6      h = 0.001,      # gradient step size
7      t = 1,          # directions averaged per estimate
8      grad_type = 'noth', # 'noth' (forward) or 'h' (central)
9      constraint_type = 'c', # 'c' (box) or 'notc' (unconstrained)
10 )
11 x_opt = opt.optimize(lower_bounds, upper_bounds)

```

Algorithm outline

At each iteration ZORM estimates the gradient of the μ -smoothed objective

$$f_\mu(\mathbf{x}) = \mathbb{E}_{\mathbf{u} \sim \mathcal{N}(0, I)}[f(\mathbf{x} + \mu \mathbf{u})]$$

by a random finite-difference estimator, then projects the gradient-descent step onto the box `[lower_bounds, upper_bounds]`. With `grad_type='noth'` (forward difference):

$$\hat{g} = \frac{f(\mathbf{x} + \mu \mathbf{u}) - f(\mathbf{x})}{\mu} \mathbf{u};$$

with `grad_type='h'` (central difference):

$$\hat{g} = \frac{f(\mathbf{x} + \mu \mathbf{u}) - f(\mathbf{x} - \mu \mathbf{u})}{2\mu} \mathbf{u}.$$

See [6] for theoretical background.

5.6 Class `Blackbox_raceline`

Blackbox_raceline optimises the lateral shift vector $\alpha \in \mathbb{R}^N$ *directly* for minimum lap time (or any user-supplied cost function). Unlike **Opt_min_CurvTime**, there is no inner QP and no curvature proxy: the true kinematic lap-time oracle is called at every function evaluation.

Motivation

Opt_min_CurvTime uses a two-stage pipeline (ZORM/CMA-ES $\rightarrow$ QP) in which the outer loop optimises spline segment lengths and the inner QP minimises curvature as a proxy for lap time. The proxy is inexact and couples geometry to the inner QP. **Blackbox_raceline** removes both stages and optimises α directly, trading inner-QP exactness for a higher cost-per-evaluation but a more faithful objective.

Mathematical formulation

$$\min_{\alpha} \text{cost}(\alpha) \quad \text{subject to} \quad \alpha_i \in [-(w_{\text{left},i} - \delta), w_{\text{right},i} - \delta] \quad \forall i,$$

where $\delta = \text{sfty}$ is the per-side safety clearance and the box bounds are identical to those used by the internal QP of **Opt_min_CurvTime**. The spline normal vectors $\hat{\mathbf{n}}_i$ are computed once from the Euclidean distances of the original reference track and held fixed throughout optimisation, so the box-projection step is exact at every iteration.

The default cost function evaluates the full kinematic lap time via the forward-backward velocity solver. Any callable `cost_fn(alpha) -> float` can be substituted.

Constructor

```

1 bb = Blackbox_raceline(
2     reftrack,          # (N,4): x, y, w_right, w_left (unclosed
3     )
4     ggv,               # GGV array (K,3) already loaded
5     ax_max_machines,   # machine limits (L,2) already loaded
6     sfty               = 1.0,   # safety half-width [m]
7     v_max              = 22.88, # maximum velocity [m/s]
8     si                 = 2.0,   # raceline interpolation step size [m]
9     fw                 = 3,     # velocity-profile filter window (None =
10    off)
11    m_veh               = 1000.0, # vehicle mass [kg]
12    drag_coeff           = 0.0,   # drag coefficient [kg/m]
13    dyn_model_exp       = 1.0,   # dynamics model exponent in [1, 2]
14    cost_fn             = None,   # callable(alpha)->float or None
15    init                 = 'random', # initialisation strategy (see below)
16    oracle               = 'gaussian', # direction oracle: 'gaussian' or '
17    sphere'
18    mu                  = 0.05,   # Z0 perturbation radius
19    h                    = 0.001, # Z0 gradient step size
20    t                    = 1,     # Z0 directions averaged per estimate
21    grad_type           = 'noth', # 'noth' (forward) or 'h' (central)
22    iterations          = 300,   # default iteration budget
23    kappa_bound         = 0.7,   # curvature bound for 'mincurv' init [1/m]
24    ]
25    seed                = None,   # global random seed
26 )

```

Initialisation modes (init).

Mode	Description
'zero'	$\alpha_0 = \mathbf{0}$ — start from the reference line
'random'	$\alpha_{0,i} \sim \text{Uniform}(\ell_i, u_i)$ — random interior point
'mincurv'	Solve the minimum-curvature QP ($\mathbf{H}_f$) and use the result as α_0 ; provides the best geometric warm-start
'shortest'	Solve the OSP QP and use the result as α_0 ; useful on high-speed tracks where path length matters more than apex curvature

Direction oracles (oracle).

- 'gaussian': $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, I_N)$
- 'sphere': $\mathbf{u}$ sampled from the unit sphere

Gradient estimators (grad_type).

- 'noth' (forward): $\hat{g} = \frac{f(\alpha + \mu u) - f(\alpha)}{\mu} u$
- 'h' (central): $\hat{g} = \frac{f(\alpha + \mu u) - f(\alpha - \mu u)}{2\mu} u$

Method: find_alpha

```

1 best_alpha, history = bb.find_alpha(
2     solver             = 'Z0',   # 'Z0' or 'CMA'
3     n_iter             = None,   # overrides bb.iterations if set
4     # Z0 parameters

```

```

5     step_size = None,      # overrides bb.h
6     mu        = None,      # overrides bb.mu
7     t         = None,      # overrides bb.t
8     grad_type = None,      # overrides bb.grad_type
9     # CMA parameters
10    sigma      = None,      # initial step size (default: mean box width
                             / 6)
11    popsize     = None,      # population size (default: 4 + 3*ln(N))
12    # Common
13    seed        = None,
14    verbose     = True,
15    print_every = 50,
16 )

```

Minimises the cost over α using either projected ZO gradient descent or CMA-ES. Both solvers maintain and return the *best-ever* alpha seen across all evaluations.

Returns:

- **best_alpha**: (N,) alpha achieving the lowest observed cost
- **history**: (n_iter+1,) best-so-far cost at each iteration/generation; **history[0]** is the cost at α_0

After the call, **bb.alpha_opt** and **bb.history** are updated.

Method: generate_raceline

```

1 s, vx, ax, kappa, t_prof, raceline_xy = bb.generate_raceline(alpha=
    None)

```

Builds the full kinematic profiles for a given α . If **alpha** is **None**, uses **bb.alpha_opt** (result of the most recent **find_alpha** call).

Returns:

Return	Description
s	(M,) cumulative arc-length [m]
vx	(M,) velocity profile [m/s] (unclosed)
ax	(M,) longitudinal acceleration [m/s ²]
kappa	(M,) curvature [m ⁻¹]
t_prof	(M+1,) cumulative lap time [s]; t_prof[-1] = total
raceline_xy	(M,2) interpolated raceline coordinates

Method: reset_alpha

```

1 bb.reset_alpha(new_init=None)

```

Re-initialises α_0 and clears **alpha_opt** and **history**. Pass **new_init** to switch the initialisation strategy; **None** repeats the current one. Useful for multi-start experiments.

Custom cost functions

Any callable **cost_fn(alpha: np.ndarray) -> float** is accepted. Even the cost function can be focused on other aspects but the lap time. The example below adds a lateral-acceleration comfort penalty to the lap time:

```

1 W_COMFORT = 0.5    # weight [s / (m/s^2)]
2
3 def comfort_cost(alpha):
4     s, vx, ax, kappa, t_prof, _ = bb.generate_raceline(alpha)
5     laptime = t_prof[-1]
6     ay_peak = float(np.max(vx**2 * np.abs(kappa))) # [m/s^2]
7     return laptime + W_COMFORT * ay_peak
8
9 bb_comfort = Blackbox_raceline(
10     reftrack, ggv, ax_max_machines,
11     init='mincurv', cost_fn=comfort_cost)
12 best_alpha, history = bb_comfort.find_alpha(solver='Z0', n_iter=200)

```

5.7 Class Clothoid_raceline

Fits a piecewise clothoid (Euler spiral) raceline where curvature varies linearly along each segment.

```

1 cl = Clothoid_raceline(
2     k_0, s, x0, y0, th0,
3     nump=10, a_max=5.3, a_min=12, ay_max=12,
4     c0=0.00002, c1=0.0015,
5     v_max=22.88, v0=0, vf=22.88,
6 )
7 x_pts = cl.X_0(a, b, c) # Fresnel-integral x displacement
8 y_pts = cl.Y_0(a, b, c) # Fresnel-integral y displacement
9 path = cl.compute_clothoid_path()

```

Fresnel-type integrals are evaluated numerically via *scipy.integrate.quad*. See [3] for the mathematical background.

6 Module: KinematicProfs.py

KinematicProfs.py computes physically consistent velocity, acceleration, and lap-time profiles along a given curvature profile, using a GGV diagram for the vehicle dynamics limits.

6.1 calc_vel_profile

```

1 vx_profile = calc_vel_profile(
2     ggv, ax_max_machines,
3     v_max, kappa, el_lengths, closed,
4     filt_window = 3,
5     drag_coeff = 0.0,
6     m_veh = 1000.0,
7     dyn_model_exp = 1.0,
8     mu = None,
9     v_start = None,
10    v_end = None,
11    p_ggv = None, # pre-tiled GGV (n,K,3); avoids
12    recomputation
13 )

```

Computes the maximum feasible velocity profile via a forward-backward integration algorithm:

1. **Lateral limit:** $v_i \leq \sqrt{a_{y,\max}(v_i)/|\kappa_i|}$

2. **Forward pass** (acceleration): integrate from a start speed while respecting $a_{x,\max}$ and the GGV ellipse.
3. **Backward pass** (deceleration): integrate backward from an end speed to enforce braking constraints.
4. **Final profile**: element-wise minimum of the three limits.

Closed-circuit periodicity (closed=True). For a closed lap the profile must satisfy $v[0] = v[-1]$ (the car enters the next lap at the same speed it left). This is achieved by a **3-lap tripling** approach:

1. The curvature array is tiled three times: $[\kappa, \kappa, \kappa]$.
2. The forward pass is run on the triple and the middle lap is extracted.
3. The backward pass is run on the middle lap result (again tripled) and the middle lap is extracted.

Tripling (rather than doubling) is necessary because the backward pass is implemented by *flipping* and running the forward algorithm. With only two laps, the first element of the flipped array is never written by the forward algorithm, leaving $v[-1]$ unconstrained and causing a large artificial deceleration spike at the lap seam. With three laps the corner at the start of the third copy constrains $v[-1]$ of the middle lap to the physically correct braking distance.

Key parameters

Parameter	Description
ggv	(K,3) array: $[v, a_{x,\max}, a_{y,\max}]$
ax_max_machines	(L,2) motor traction table: $[v, a_{x,\text{drive}}]$
v_max	Hard speed cap [m/s]
kappa	Curvature array $[\text{m}^{-1}]$, length n
el_lengths	Segment lengths [m], length n
closed	True for a closed lap (3-lap tripling used)
filt_window	Moving-average filter window; None disables filtering
drag_coeff	Aerodynamic drag [kg/m]; effective deceleration $a_{\text{drag}} = c_{\text{drag}} v^2 / m$
m_veh	Vehicle mass [kg]
dyn_model_exp	GGV ellipse exponent in $[1, 2]$; 1 = diamond, 2 = ellipse
v_start	Start velocity [m/s] for the unclosed case
p_ggv	Pre-tiled (n,K,3) GGV array; skips per-call tiling

6.2 calc_ax_profile

```

1 ax_profile = calc_ax_profile(
2     vx_profile, el_lengths, eq_length_output=False)

```

Numerically differentiates the velocity profile to give the longitudinal acceleration $a_x = (v_{i+1}^2 - v_i^2) / (2 \Delta s_i)$ at each point. Input `vx_profile` should be the closed version (length $n + 1$, last element = $v[0]$).

6.3 calc_t_profile

```

1 t_profile = calc_t_profile(
2     vx_profile, el_lengths, t_start=0.0, ax_profile=None)

```

Integrates the velocity profile to give the elapsed time at each sample point. Returns a vector of length $n + 1$; the last entry is the total lap time.

6.4 `curvature_profile` and `curvature_profile2`

```

1 kappa = curvature_profile(points)      # via scipy.interpolate.splprep
2 kappa = curvature_profile2(reftrack)   # spline-derivative method

```

Two alternative functions for computing the curvature of a discrete point sequence. **`curvature_profile`** uses *scipy.interpolate.splprep*; **`curvature_profile2`** uses the raw spline derivatives directly.

6.5 `cumulative_distances`

```

1 s = cumulative_distances(el_lengths)

```

Returns the cumulative arc-length from the first point, starting at zero. Equivalent to `np.concatenate([[0], np.cumsum(el_lengths)])`.

7 Module: `opt_mintime.py`

`opt_mintime` solves a full minimum-lap-time optimal-control problem (OCP) formulated in the curvilinear frame of the track. It uses CasADi to construct the nonlinear programme (NLP) symbolically and IPOPT as the NLP solver. See [2] for the full mathematical formulation.

Function signature

```

1 (alpha_opt, v_opt, raceline_opt,
2  reftrack_interp_out, a_interp_tmp,
3  normvec_out, s_opt, laptime) = opt_mintime(
4      reftrack,          # (n,4): x, y, w_right, w_left
5      coeffs_x,          # (n,4) spline x-coefficients
6      coeffs_y,          # (n,4) spline y-coefficients
7      normvectors,       # (n,2) unit normals
8      pars,              # dict of vehicle and solver parameters
9      print_debug = False,
10     plot_           = False,
11 )

```

Vehicle and tyre parameters are supplied via the `pars` dictionary, which is typically loaded from `params/racecar.ini`. Key sub-dicts include `vehicle_params_mintime`, `tire_params_mintime`, and `optim_opts`.

8 Usage Examples

The following two scripts, located in the `tests/` directory, provide complete end-to-end demonstrations of the package.

8.1 General_test.py – Geometric, Proxy, and Blackbox Optimisation

Run from the `tests/` directory:

```
cd tests && python General_test.py
```

Stage 1 – Track import and preprocessing

Loads the Catalunya centreline CSV, sub-samples to reduce computation time, and imports the GGV and `ax_max_machines` tables.

```
1 MAP_PATH = "maps/Catalunya/Catalunya_centerline.csv"
2 csv_data = np.loadtxt(MAP_PATH, comments='#', delimiter=',')
3 reftrack_full = csv_data[:, 0:4] # [x, y, w_right, w_left]
4 reftrack = subsample_track(reftrack_full, step=4)
5 ggv, ax_max_machines = import_veh_dyn_info(GGV_PATH, AX_MAX_PATH)
```

Stage 2 – Spline computations

Fits C2-continuous cubic splines through the reference points, recovers segment lengths, interpolates at a uniform 2 m step, and computes analytical and numerical heading and curvature profiles.

```
1 coeffs_x, coeffs_y, M, normvec_norm = calc_splines(
2     path=np.vstack((reftrack[:, :2], reftrack[0, :2])),
3     el_lengths=ds_0, use_dist_scaling=True)
4 spline_lengths = calc_spline_lengths(coeffs_x, coeffs_y)
5 path_interp, spline_inds, t_vals, s_pts = interp_splines(
6     coeffs_x, coeffs_y, spline_lengths, stepsize_approx=2.0)
7 psi_an, kappa_an = calc_head_curv_an(coeffs_x, coeffs_y,
8                                     spline_inds, t_vals)
```

Stage 3 – Minimum-curvature raceline

Solves the curvature-minimisation QP, then reconstructs the raceline and computes velocity and lap-time profiles.

```
1 alpha_mincurv, curv_error = solvers.opt_min_curv(
2     reftrack=reftrack, normvectors=normvec_norm, A=M,
3     kappa_bound=0.5, w_veh=1.0)
4 raceline_mc, *_ = create_raceline(
5     reftrack[:, :2], normvec_norm, alpha_mincurv, stepsize_interp
6     =2.0)
7 vx_mc = calc_vel_profile(ggv, ax_max_machines, v_max=22.88,
8                           kappa=kappa_mc, el_lengths=el_mc, closed=
9     True,
10                        drag_coeff=0.75, m_veh=1000)
11 t_mc = calc_t_profile(vx_mc, ax_mc, el_mc)
```

Stage 4 – Shortest-path raceline

Solves the shortest-path QP and returns the full raceline and kinematic profiles.

```
1 (raceline_sp, alpha_sp, s_sp,
2  vx_sp, ax_sp, kappa_sp, t_sp) = solvers.ShortestPath(
3     reftrack=reftrack, w_veh=1.0, stepsize=2.0, plot=False,
4     ggv_import_path=GGV_PATH,
```

```
5 ax_max_machines_import_path=AX_MAX_PATH)
```

Stage 5 – Kinematic profiles

Demonstrates **KinematicProfs** in isolation using the minimum-curvature raceline from Stage 3: plots velocity, longitudinal acceleration, and curvature against arc length.

Stage 6 – Full optimisation comparison (ZO vs CMA-ES)

Instantiates **Opt_min_CurvTime** and benchmarks ZORM, CMA-ES, minimum-curvature, and centre-line solutions.

```
1 optct = solvers.Opt_min_CurvTime(
2     reftrack=reftrack, center=reftrack, mu=0.01, h=0.001,
3     kapb=0.5, sfty=1.0, si=2, vm=22.88, m_veh=1000,
4     drag_coeff=0.75, MC=1, sigma=0.005,
5     iterations_ZO=100, iterations_CMA=10, popsize=16,
6     ggv_import_path=GGV_PATH,
7     ax_max_machines_import_path=AX_MAX_PATH, fw=3)
8
9 raceline_zo, ds_zo = optct.generate_raceline(solver='ZO')
10 s_zo, vx_zo, ax_zo, kappa_zo, t_zo, rl_zo = \
11     optct.generate_kinProfs(ds=ds_zo)
12 s_opt, vx_opt, ax_opt, kappa_opt, t_opt, rl_opt = \
13     optct.Comparison(plot='Y', output='ZO')
```

Stage 7 – In-loop path refinement

Demonstrates the **refine_every** feature of **Opt_min_CurvTime**. The total iteration budget is split into blocks of q iterations; after each block (except the last) the current best raceline is used to rebuild the reference track, and optimisation continues on the geometrically tighter track. Three variants are compared head-to-head with the same total budget and the same optimizer instance; a call to **reset()** restores the original track between variants.

```
1 REFINE_Q_ZO    = 33    # ZO cadence: 100 iters -> 3 blocks, 2
   refinements
2 REFINE_Q_CMA   = 3     # CMA cadence: 10 iters -> 3 blocks, 2
   refinements
3
4 optct_ref = solvers.Opt_min_CurvTime(
5     reftrack=reftrack, center=reftrack,
6     mu=0.01, h=0.001, kapb=0.5, sfty=1.0, si=2,
7     vm=22.88, m_veh=1000, drag_coeff=0.75, MC=1,
8     sigma=0.005, iterations_ZO=100, iterations_CMA=10,
9     popsize=16, fw=3,
10    ggv_import_path=GGV_PATH,
11    ax_max_machines_import_path=AX_MAX_PATH,
12    refine_every=REFINE_Q_ZO,    # instance default for ZO calls
13    refine_subsample=1,          # keep all refined control points
14 )
15
16 # A: ZO without refinement (override refine_every=0)
17 s_a, vx_a, ax_a, kappa_a, t_a, rl_a = optct_ref.generate_kinProfs(
18     solver='ZO', refine_every=0)
19 optct_ref.reset()
20
21 # B: ZO with in-loop refinement (uses instance default refine_every)
```

```

22 s_b, vx_b, ax_b, kappa_b, t_b, rl_b = optct_ref.generate_kinProfs(
23     solver='ZO')
24 optct_ref.reset()
25
26 # C: CMA-ES with in-loop refinement (cadence overridden per-call)
27 s_c, vx_c, ax_c, kappa_c, t_c, rl_c = optct_ref.generate_kinProfs(
28     solver='CMA', refine_every=REFINE_Q_CMA)
29 optct_ref.reset()

```

The output includes a results table and a two-panel figure: the left panel overlays the three racelines on the track boundaries; the right panel compares the velocity profiles.

Variant	Lap time [s]
A. ZO – no refinement (100 iters)	t_A
B. ZO – refine every 33 iters (100 iters)	t_B
C. CMA-ES – refine every 3 iters (10 iters)	t_C

Note on refinement behaviour. After each refinement `self.reftrack` is replaced by the current best raceline and the segment-length bounds (`min_s`, `max_s`) are recomputed from the new control-point spacing. With small per-block budgets the ZO optimiser does not fully converge before the next refinement resets the search landscape. Call `reset()` at any time to restore the original track.

Stage 8 – Blackbox direct-alpha optimisation

Demonstrates `Blackbox_raceline` with all four initialisation strategies and both solvers, then shows a custom comfort-aware cost function. The stage produces convergence-history plots, per-init improvement comparisons (raceline maps, velocity profiles, Δv_x profiles), and alpha-shift profiles.

```

1  BB_SFTY = 0.2;  BB_ITERS_ZO = 450;  BB_ITERS_CMA = 30
2
3  # --- 8a: initial cost comparison ---
4  for mode in ['random', 'zero', 'mincurv', 'shortest']:
5      bb = solvers.Blackbox_raceline(
6          reftrack=reftrack, ggvs=ggvs, ax_max_machines=ax_max_machines,
7          sfty=BB_SFTY, v_max=22.88, si=2.0, init=mode)
8      t0 = bb._eval_cost(bb.alpha_0)
9      print(f" {mode}: laptime0 = {t0:.4f} s")
10
11 # --- 8b: ZO on all inits ---
12 for mode in ['random', 'zero', 'mincurv', 'shortest']:
13     bb_zo = solvers.Blackbox_raceline(
14         reftrack=reftrack, ggvs=ggvs, ax_max_machines=ax_max_machines,
15         sfty=BB_SFTY, v_max=22.88, si=2.0, init=mode,
16         mu=0.001, h=0.0001, t=2)
17     best_a, hist = bb_zo.find_alpha(
18         solver='ZO', n_iter=BB_ITERS_ZO, verbose=True)
19
20 # --- 8c: CMA-ES on mincurv and shortest ---
21 for mode in ['mincurv', 'shortest']:
22     bb_cma = solvers.Blackbox_raceline(..., init=mode)
23     best_a, hist = bb_cma.find_alpha(
24         solver='CMA', n_iter=BB_ITERS_CMA,
25         sigma=0.1, popsize=20)
26

```

```

27 # --- 8d: custom cost function ---
28 def comfort_cost(alpha):
29     s, vx, ax, kappa, t_prof, _ = _bb_ref.generate_raceline(alpha)
30     return t_prof[-1] + 0.5 * np.max(vx**2 * np.abs(kappa))
31
32 bb_custom = solvers.Blackbox_raceline(
33     ..., init='shortest', cost_fn=comfort_cost)
34 best_custom, hist_custom = bb_custom.find_alpha(
35     solver='Z0', n_iter=BB_ITERS_Z0)
36
37 # --- 8e/8f: profiles and plots ---
38 s, vx, ax, kappa, t_prof, raceline = bb_zo.generate_raceline()
39 print(f"Z0 laptime: {t_prof[-1]:.3f} s")

```

8.2 General_test_optmintime.py – Minimum Lap-Time Optimal Control

```
cd tests && python General_test_optmintime.py
```

Stage 1 – Track import

Loads and sub-samples the centreline identically to `General_test.py`.

Stage 2 – Configuration loading

Parses `params/racecar.ini` with `configparser` to populate vehicle and solver parameter dictionaries, and loads the GGV table.

```

1 parser = configparser.ConfigParser()
2 parser.read("params/racecar.ini")
3 pars["vehicle_params_mintime"] = json.loads(
4     parser.get('OPTIMIZATION_OPTIONS', 'vehicle_params_mintime'))
5 if pars["optim_opts"]["ax_pos_safe"] is None:
6     pars["optim_opts"]["ax_pos_safe"] = np.amin(ggv[:, 1])

```

Stage 3 – Track preprocessing

Calls `utils.prep_track` to smooth and re-sample the centreline at the collocation stepsize and to compute spline coefficients and unit normals.

```

1 reftrack_interp, normvec_normalized_interp, a_interp, \
2     coeffs_x_interp, coeffs_y_interp = utils.prep_track(
3     reftrack_imp=reftrack,
4     reg_smooth_opts=pars["reg_smooth_opts"],
5     stepsize_opts=pars["stepsize_opts"],
6     debug=True)

```

Stage 4 – Minimum lap-time optimisation (pass 1)

Solves the full OCP via Gauss–Legendre collocation and IPOPT.

```

1 (alpha_opt, v_opt, raceline_opt, reftrack_interp_out,
2  a_interp_tmp, normvec_out, s_opt, laptime) = opt_mintime.opt_mintime(
3     reftrack=reftrack_interp,
4     coeffs_x=coeffs_x_interp, coeffs_y=coeffs_y_interp,
5     normvectors=normvec_normalized_interp,
6     pars=pars, print_debug=True, plot_=True)

```

Stage 5 – Re-optimisation pass (widened corridor)

Widens the optimisation corridor and re-solves the NLP using the pass-1 track geometry as the new reference.

```

1 pars_reopt = copy.deepcopy(pars)
2 pars_reopt["optim_opts"]["width_opt"] = w_veh_widened
3 alpha_reopt, v_reopt, *_ , laptime_reopt = opt_mintime.opt_mintime(
4     reftrack=reftrack_interp_out, normvectors=normvec_out,
5     pars=pars_reopt, ...)
6 print(f"Pass 1: {laptime:.2f} s    Pass 2: {laptime_reopt:.2f} s")

```

9 GGV Diagram Format

The GGV file must be comma-separated with **no header row**:

Column 1	Column 2	Column 3
v [m/s]	$a_{x,\max}$ [m/s ²]	$a_{y,\max}$ [m/s ²]
0.0	10.0	12.0
10.0	10.0	11.5
20.0	9.5	10.0
$\vdots$	$\vdots$	$\vdots$

The `ax_max_machines` file has two columns: $[v, a_{x,\text{drive}}]$. Both files must be monotone in velocity and cover the full range up to `v_max`.

10 Track Map Format

Track CSV files must have **four columns** and **no header row**:

Column 1	Column 2	Column 3	Column 4
x [m]	y [m]	w_{right} [m]	w_{left} [m]
0.00	0.00	5.0	5.0
5.23	0.12	5.0	5.0
$\vdots$	$\vdots$	$\vdots$	$\vdots$

Sign convention: w_{right} is the distance from the reference line to the right boundary (measured along the outward normal); w_{left} is the distance to the left boundary. Both are positive.

The track is assumed **closed**: the last point connects back to the first. Do *not* repeat the first point at the end of the file.

11 Dependencies and Licences

Library	Licence	URL
NumPy	BSD-3-Clause	https://numpy.org
SciPy	BSD-3-Clause	https://scipy.org
Matplotlib	PSF/Matplotlib	https://matplotlib.org
quadprog	GPL-2.0	https://github.com/quadprog/quadprog
CasADi	LGPL-3.0	https://web.casadi.org
IPOPT	EPL-2.0	https://coin-or.github.io/Ipopt

11 References

- [1] Francesco Braghin, Federico Cheli, Stefano Melzi, and Edoardo Sabbioni. Race driver model. *Computers & Structures*, 86(13-14):1503–1516, 2008.
- [2] Fabian Christ, Alexander Wischnewski, Alexander Heilmeier, and Boris Lohmann. Time-optimal trajectory planning for a race car considering variable tyre-road friction coefficients. *Vehicle system dynamics*, 59(4):588–612, 2021.
- [3] Marco Frego, Enrico Bertolazzi, Francesco Biral, Daniele Fontanelli, and Luigi Palopoli. Semi-analytical minimum time solutions with velocity constraints for trajectory following of vehicles. *Automatica*, 86:18–28, 2017.
- [4] Nikolaus Hansen. The cma evolution strategy: A tutorial. *arXiv preprint arXiv:1604.00772*, 2016.
- [5] Alexander Heilmeier, Alexander Wischnewski, Leonhard Hermansdorfer, Johannes Betz, Markus Lienkamp, and Boris Lohmann. Minimum curvature trajectory planning and control for an autonomous race car. *Vehicle System Dynamics*, 2020.
- [6] Yurii Nesterov and Vladimir Spokoiny. Random gradient-free minimization of convex functions. 17(2):527–566.